ByteWise

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MCPs, Skills and Plugins

How MCP servers, skills and plugins turn your AI agent from a chat buddy into a Power BI teammate.



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THE AGENT, EXPLAINED

Brains, hands and a laptop

The owl is the model: it does the thinking. Here's what it knows and how it gets things done.

THE MANUALS

Skills + instructions

A shelf of skills and instructions. It opens only the one the task needs.

SKILL.md

BONUS · PLUGINS

The backpack: skills, MCP and hooks, packed together.

RIGHT HAND · CLI

Types in the terminal

Runs fab, te, dscmd with your permissions.

> fab ls

LEFT HAND · MCP

Plugs into live systems

Typed tools, discovered at runtime. No shell needed.



INTERFACE · HARNESS · MODEL

Three separate choices

1 · INTERFACE

Where you work

Chat view or Agents window: where you type, watch and review.

2 · HARNESS

What runs the agent

Copilot, Claude, Codex or Local. Decides customizations, hook formats and permissions.

3 · MODEL

What does the thinking

Whatever you pick in the model picker. The owl from slide 2.

⚠ Common mix-ups

- A Claude **model** inside Copilot still runs on the Copilot **harness**
- "Local" is the name of an older harness, not "runs on your machine"
- Supported sessions can move between the Chat view and the Agents window

Source: code.visualstudio.com/docs/agents/run/agent-harnesses

GITHUB COPILOT IN VS CODE

Shape the agent: guide it with instructions, equip it with skills

LAYER	USER LEVEL · follows you	PROJECT LEVEL · shared via repo
Instructions Guide · always-on or scoped context	~/.copilot/copilot-instructions.md and ~/.copilot/instructions/ Local agent: VS Code profile, roams with Settings Sync	.github/copilot-instructions.md or AGENTS.md *.instructions.md scoped by **applyTo** glob or **description**
Prompt files Trigger · reusable requests you run with /name	VS Code profile, roams with Settings Sync Local agent only	.github/prompts/*.prompt.md Deprecated for Agent Host: migrate to skills
Skills Equip · SKILL.md + scripts, loaded on demand	~/.copilot/skills/, ~/.agents/skills/, ~/.claude/skills/ Personal toolbox, available in every repo	.github/skills/<name>/ (also .agents, .claude) Open standard: same skill runs in VS Code, CLI, cloud agent

User + project (+ org) instructions are additive, not overriding · code.visualstudio.com/docs/agent-customization

GITHUB COPILOT IN VS CODE

Five layers, two scopes, one package

LAYER	USER LEVEL · follows you	PROJECT LEVEL · shared via repo
Instructions Guide · always-on or scoped	~/.copilot/copilot-instructions.md ~/.copilot/instructions/	.github/copilot-instructions.md, AGENTS.md .github/instructions/*.instructions.md
Prompt files Trigger · /name, Local only	VS Code profile (Local agent)	.github/prompts/*.prompt.md
Custom agents Who · role, tools, model	~/.copilot/agents/*.agent.md	.github/agents/*.agent.md
Skills Equip · loaded on demand	~/.copilot/skills/, ~/.agents/skills/	.github/skills/<name>/SKILL.md
MCP servers Reach · external tools, data	~/.copilot/mcp-config.json	.mcp.json (portable) or .vscode/mcp.json

Plugins Pack skills, agents, hooks and MCP servers into one installable unit

CLIs: Copilot CLI shares many of these portable formats; tools like fab run in the terminal

SKILLS · SKILL.MD

Equip it: skills add up and load on demand

HOW A SKILL LOADS

1 Discover
Only **name** and **description** are read up front

2 Load
The SKILL.md body enters context when the request matches

3 Use
Scripts and references load only if SKILL.md points to them

Two ways in

Automatically by description match, or explicitly as /skill-name. Frontmatter can limit either one.

HOW THEY COMBINE

User + project = one pool

No inheritance, no override: every discovered skill is a candidate. Available does not mean guaranteed to be used.

⚠ Attention point

- **Same name** in two places: both listed, no general winner (multi-root: primary root wins)
- **Similar descriptions**: the agent may pick the wrong skill, or none
- Keep names unique, e.g. pbip-conventions vs my-pbip-shortcuts
- Write each description as a trigger: "Use when..." with distinct keywords

Same skill folder works in VS Code, Copilot CLI and cloud agent · Source: code.visualstudio.com/docs/agent-customization/agent-skills

PROMPT FILES · .PROMPT.MD

Prompt files vs skills: who pulls the trigger

Skill: the agent picks it up

- Loads when the description matches, or when you call /name
- A capability: bundles scripts, templates and references
- Progressive loading keeps the context small
- Open standard: VS Code, Copilot CLI, cloud agent

e.g. pbip-conventions

Prompt file: you run it

- Start it with /name, Chat: Run Prompt or the play button
- One reusable request; frontmatter can pick agent, model, tools
- Asks for input with ${input:name}, reuses ${selection}
- Lives in .github/prompts or your VS Code profile

e.g. /review-measures

⚠ **Heads-up:** Agent Host sessions don't load prompt files, and VS Code recommends migrating them to skills.
They still work with the Local agent, for now.

code.visualstudio.com/docs/agent-customization/prompt-files

HOOKS · PREVIEW

Enforce it: your code runs at fixed points in the agent's lifecycle

Where hooks live

Project · .github/hooks/*.json
User · ~/.copilot/hooks/*.json
Custom agent · hooks: in .agent.md frontmatter (Local only)
Plugin · hooks.json shipped with the plugin

Keep in mind

- The model can't skip a hook, but hooks only run when enabled and configured, and some failures just warn.
- The harness decides the format: Copilot uses the SDK/CLI schema, Local uses VS Code's.
- Hooks run with your permissions. Review them like any other code.

LIFECYCLE EVENTS · typical use

SessionStart	Inject project context
UserPromptSubmit	Audit or enrich requests
PreToolUse	Block, ask approval, rewrite input
PostToolUse	Run formatter or validator
PreCompact	Save state before compaction
SubagentStart	Track or brief subagents
SubagentStop	Check subagent results
Stop	Verify done, or force another turn

Events and locations shown are for VS Code's Local harness · code.visualstudio.com/docs/agent-customization/hooks

MCP · MODEL CONTEXT PROTOCOL

Reach it: MCP plugs the agent into live systems

CLIENT

Your agent

Copilot in VS Code asks: what tools do you have?

MCP SERVER

Typed tools

Names, inputs, descriptions: discovered at runtime

LIVE SYSTEM

Model, workspace

Desktop, Fabric, docs: whatever the server reaches

Local server

Runs on your machine, started by the agent (stdio).

`npx ...powerbi-modeling-mcp`

Reaches local files and Desktop.

Hosted server

A URL you sign in to (HTTP). Nothing to install.

Fabric IQ, Fabric Core

Acts as you, with your rights.

Set up and control

`.mcp.json`

`~/.copilot/mcp-config.json`

You approve tool calls; agents can be limited to server/*.

code.visualstudio.com/docs/agent-customization/mcp-servers,

CLI · COMMAND-LINE TOOLS

Type it: the agent uses the same tools you do

HOW IT WORKS

1 Write the command
The agent picks a tool and its flags

2 Run it in the terminal
As you, with your sign-ins and rights

3 Read the output
Then fix, retry or move on

fab · te · dscmd.exe · git

WHAT TO KNOW

How it knows the command

- From training: fine for git, risky for new tools
- From --help: slower, but current
- From a skill: exact and repeatable

Watch out

- It acts with your permissions
- Review commands before you approve
- Guessed names fail: dscmd, not daxstudio

MCP VS CLI

Two routes to the same capability



Where they meet: an MCP server often wraps the very same CLI or API, and Claude Code is a CLI that hosts MCP.

</> **Shell** = the command interpreter inside a terminal (PowerShell, bash). An agent with a shell types CLI commands just as you would, with your permissions. No way to run commands, no CLI route.

MCP VS CLI

When to reach for which

Aspect	CLI	MCP
Needs a shell	Yes	No: works in chat apps and IDEs
How usage is learned	Training, --help, or a skill	Schemas delivered automatically
Context cost	Low until used	Tool schemas use context; tool search can trim it
Composability	Pipes and scripts, no model in between	Each result returns to the model
Access control	OS + service permissions	Client approvals + server/service authorization
Best for	Mature tools, bulk work	Live systems, governed access



Takeaway: a CLI is a door built for people that agents can also walk through. MCP is a door built for agents.

Ship it: one install, the whole toolkit

EXAMPLE · SHOWCASE REPO

```
bytewise-pbip/
  plugin.json          # name, version
  skills/
    pbip-conventions/  # the manual
  mcp.json             # the hands
  com.github.copilot/
    agents/
    hooks/
```

Microsoft ships its own: the Power BI authoring plugin in skills-for-fabric.

What goes in

- Portable: skills and MCP servers
- Copilot-specific: agents, hooks, commands

How you install it

- From a marketplace, in the Plugins page
- From a Git repo: Install Plugin From Source
- From a local folder, while you build it

Before you install

Hooks and MCP servers run code on your machine. Review a plugin like any other code.

code.visualstudio.com/docs/agent-customization/agent-plugins

FABCON BARCELONA 2026 · ANNOUNCEMENTS

Read the status before you pick the tool

Announcement	Status	Status distinction	Impact on tool selection
Power BI agentic experiences and PBIP go GA	GA	Local authoring MCP, authoring skills and Desktop Bridge included	PBIP/TMDL/PBIR + source control + agent verification becomes the stronger default
Hosted Power BI Authoring MCP	Preview	Preview, even though local authoring is GA	Authoring models in the cloud no longer needs a local modeling server
Fabric IQ in Copilot Chat and Cowork	GA	GA there; Code integration coming via Frontier	Governed semantics now serve agents beyond reports. Not the same as model-edit rights
Apache Ossie and Microsoft converter	Early	Early implementation, not a TOM/TMDL replacement	A real interoperability CLI to watch; limited on richer Power BI constructs
Agentic app creation / Fabric Apps expansion	Preview	Expanding to Pro/PPU, rolling out, not universal GA	Pushes toward app code, SQL writeback and ops workflows. No new standalone "Fabric Apps MCP"

GETTING STARTED · PREREQUISITES

Before you plug anything in

1 · The basics

- VS Code + GitHub Copilot, in agent mode
- Git, for PBIP files and repo skills
- Power BI Desktop (Windows), saved as PBIP
- A deep-reasoning model: quality depends on it

2 · Runtimes, only if you use the tool

- Node.js 22+: Copilot CLI, npx MCPs, Bridge CLI
- Python 3.10–3.13: pip install ms-fabric-cli
- Azure CLI: az login for the local Fabric MCP
- Hosted MCPs: nothing to install, just sign in

3 · Access and settings

- GitHub: "MCP servers in Copilot" policy on
- Fabric models: XMLA Read Write + Write/Build
- Hosted Authoring MCP: tenant setting on
- Fabric IQ: work account, delegated sign-in

4 · Managed laptops and gotchas

- Desktop Bridge: admins can switch it off
- Windows: PowerShell 6+ for Copilot CLI
- npm ignore-scripts=true breaks the CLI install
- Mac: no local Authoring MCP, use hosted

Power BI: one MCP to consume, one to author

Server	Hosting	What it enables	Status	Watch out
Fabric IQ MCP	MS-hosted	Find reports and models, inspect schemas, run DAX	GA	Read-only, honours RLS/OLS; user sign-in only, no service principal
powerbi-modeling-mcp (Authoring, local)	Local	Edit models, measures, roles; validate DAX; PBIP/TMDL	GA	Transactions, traces; Fabric edits need XMLA R/W
Power BI Authoring (hosted)	MS-hosted	Author Fabric semantic models, no install	Preview	No local PBIP or transactions; don't register alongside local
Power BI Consumption	MS-hosted	Query existing semantic models	Preview	Older endpoint; use Fabric IQ for new work
Microsoft Learn MCP	MS-hosted	Search docs, fetch articles and code samples	Available	Free, no auth; docs only, no tenant access

Status as of Oct 2026, check each product page · Server names link to docs · Fabric MCP server directory

Fabric platform: manage, query, orchestrate

Server	Hosting	What it enables	Status	Watch out
Fabric Core MCP	MS-hosted	Catalog search; manage workspaces, items, roles; list capacities	Available	Management, not data queries; can be destructive
Fabric MCP Server (local)	Local	API specs, item schemas, OneLake files, Data Factory	Open source	Docs tools work offline; live ops need auth
Fabric Data Warehouse MCP	MS-hosted	Run T-SQL on Warehouses and SQL endpoints	Preview	Global or item-scoped; SQL permissions apply
DataFactory MCP	Local	Manage connections, dataflows, pipelines	Available	Pipelines in preview, need the --pipeline flag
SQL MCP Server (Data API builder)	Self-hosted	Expose chosen DB entities, incl. SQL database in Fabric	DAB 1.7+	Configurable; not the Warehouse endpoint
Fabric GraphQL MCP sample	Self-built	Discover a GraphQL schema, run queries	Sample	Tutorial code, not a hosted endpoint

Status as of Oct 2026, check each product page · Server names link to docs · Fabric MCP server directory

Real-Time Intelligence and agents

Server	Hosting	What it enables	Status	Watch out
Real-Time Intelligence MCP (local)	Local	Eventhouse, Eventstreams, Activator, maps; ADX	Preview	Open source; broader than Eventhouse-only
Eventhouse MCP	MS-hosted	Discover KQL schemas, generate and run queries	Preview	Schema fetch needs Fabric Copilot enabled
Activator MCP	MS-hosted	Create, start, stop rules with Teams or email actions	Preview	KQL-backed, per item; narrower than Activator
Operations agent MCP	MS-hosted	Agent instructions, playbooks, monitoring	Preview	Part of the RTI MCP preview family
Fabric data agent as MCP	Hosted, per agent	Answers grounded in the agent's configured sources	Needs agent	Returns an answer, not query tools like Fabric IQ
Ontology MCP	Hosted, per ontology	Business entities, definitions, relationships	Preview	Needs capacity and tenant settings; SPN issue noted

Status as of Oct 2026, check each product page · Server names link to docs · Fabric MCP server directory

Fabric platform: provision, manage, move

CLI	Owner	Main use	Watch out
Fabric CLI · fab	Microsoft	Workspaces and items, import/export, pipelines, notebooks, OneLake files, CI/CD	Open source, package ms-fabric-cli. The general-purpose Fabric CLI
Azure CLI · az fabric	Microsoft	Create, resize, suspend, resume and inspect capacities	GA extension. Capacity management, not a replacement for fab
Terraform + Fabric provider	HashiCorp CLI, Microsoft provider	Declaratively provision workspaces, items, connections, role assignments	Provider is not covered by Microsoft/Fabric support policies
AzCopy · azcopy	Microsoft	Bulk upload, download and copy data to and from OneLake	Moves data and files, not item or workspace definitions

CLI names link to their docs · Check versions and licences before adopting

SQL in Fabric: deploy, transform, query

CLI	Owner	Main use	Watch out
SqlPackage · sqlpackage	Microsoft	DACPAC/BACPAC extract, import/export, schema deploy for SQL database in Fabric	Support varies by SQL workload: Warehouse is not SQL database
dbt Core + dbt-fabric	dbt Labs CLI, Microsoft adapter	SQL transformations, tests, docs and deployment on Fabric Warehouse	Check auth and driver needs for your adapter version
SQLMesh + Fabric adapter	Tobiko Data, community adapter	SQL transformation workflows on Fabric Warehouse	Limited community support; don't use Warehouse as state store
fabric-dw · fdw	Sam Debruyn (community)	Fabric SQL querying, administration, optimization, security	Community CLI; also exposes MCP functionality

Power BI Desktop and semantic models

CLI	Owner	Main use	Watch out
Desktop Bridge CLI · powerbi-desktop	Microsoft	Find Desktop instances, open reports, reload changes, screenshot pages	Windows, local Desktop only; the bridge is not an MCP
Tabular Editor 2 · TabularEditor.exe	Tabular Editor	C# scripting, Best Practice Analyzer, serialization, XMLA deploy	Free, open source, Windows. A proven CI/CD option
Tabular Editor CLI · te	Tabular Editor	Cross-platform model editing, validation, deploy, refresh, tests	Limited preview, expires 31 Oct 2026; not recommended for production CI/CD
DAX Studio · dscmd.exe	DAX Studio project	Scripted DAX runs, result export, VPAX automation	The command is dscmd.exe, not "daxstudio"

CLI names link to their docs · Check versions and licences before adopting

Report files and interoperability

CLI	Owner	Main use	Watch out
pbi-tools / pbi-tools.core	Community	PBIX/PBIT extract, compile, convert; source control and deployment	Desktop edition needs Windows; compile limits vary by artifact
pbir.tools · pbir	Max Anatsko, Kurt Buhler	Create, edit, validate PBIR; bindings, themes, filters; download and publish	Paid consulting use needs written permission; no derivative works
Apache Ossie converter · ossie-microsoft	Apache, Microsoft contributors	CLI: BIM/TMSL to and from Ossie JSON/YAML; TMDL at library level	Early stage; unsupported constructs need warnings; not lossless

Agent tip: wrap a CLI in a skill script with the exact command and flags, so the agent runs it instead of guessing